

Topology PhD Exam, August 2024

Work the following problems and show all work. Support all statements to the best of your ability.

1. Let $X = \mathbb{R}/\mathbb{Z}$ be the quotient space obtained from the real line $\mathbb{R}$ by collapsing the set of integers to a point.
 - (a) Is X metrizable?
 - (b) Does X admit a CW complex structure?
 - (c) Compute the fundamental group of X .
 - (d) Compute homology groups of X .
2. This problem has three parts.
 - (a) Prove that the 3-dimensional real projective space $\mathbb{R}P^3$ is orientable.
 - (b) Does there exist a degree 2 continuous map $f : \mathbb{R}P^3 \rightarrow T^3$ where $T^3 = S^1 \times S^1 \times S^1$ is a 3-dimensional torus?
 - (c) Does there exist a degree 2 continuous map $f : T^3 \rightarrow \mathbb{R}P^3$?
3. Let $A \subset \mathbb{R}^2$ be an infinite countable subspace.
 - (a) Can A be connected?
 - (b) Is $\mathbb{R}^2 \setminus A$ connected?
4. Does there exist a closed orientable n -manifold M with
 - (a) $n = 4$ and $H_2(M) = \mathbb{Z}$?
 - (b) $n = 5$ and $H_3(M) = \mathbb{Z}$?
 - (c) $n = 6$ and $H_3(M) = \mathbb{Z}$?
5. Prove that the fundamental group of a topological group is abelian.

Answer the following with complete definitions or statements or short proofs.

6. Compute the Euler characteristic of the manifold $M = \mathbb{H}P^3 \times \mathbb{C}P^4 \times \mathbb{R}P^5$.
7. State the Lefschetz Fixed Point Theorem. Derive the Brouwer Fixed Point Theorem from the Lefschetz Fixed Point Theorem.
8. State the Baire Category Theorem.
9. Suppose that A is a retract of X . Show that $H_i(A)$ is a direct summand of $H_i(X)$.
10. Give the definition of the mapping cylinder. Is the Möbius band a mapping cylinder?
11. Is the 2-sphere S^2 with k points removed homeomorphic to a topological group for
 - (a) $k = 1$?
 - (b) $k = 2$?

(c) $k = 3$?

12. Give the definition of the mapping cone. Is there a map $f : X \rightarrow Y$ with the mapping cone C_f homeomorphic to $\mathbb{R}P^3$?

13. Which of the following spaces are topological groups?

(a) the torus T^2 ;

(b) the Klein bottle K ;

(c) the projective space $\mathbb{R}P^3$.

14. Is $\mathbb{R}^\omega$ connected in the uniform topology?

15. State the Five Lemma.